

1) Calculate resistance in ohms

Circuit a

$$R_T = R_1 + R_2 + R_3 \\ = 20\Omega + 30\Omega + 50\Omega = 100\Omega$$

Circuit b

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \\ = \frac{1}{20} + \frac{1}{30} + \frac{1}{50}$$

$$\frac{1}{R_T} = 0.08$$

$$R_T = 12.5\Omega$$

ii) Total current from power supply

Circuit a

$$\text{Ohms law} = I = V / R_{T \text{ total}} \\ = 125V / 100\Omega \\ = 1.25A$$

Circuit b

- in parallel voltage across

$$I_{\text{total}} = 125V / 12.5\Omega \\ = 10A$$

iii) Current thru each resistor

Circuit a

- in series current is the same throughout hence!

$$I_1(20\Omega) = I_2(30\Omega) = I_3(50\Omega) = 1.25A$$

Circuit b

- parallel voltage across resistor is the same (125V)

$$I_1 (20\Omega) = 125V \div 20\Omega = 6.25A$$

$$I_2 (30\Omega) = 125V \div 30\Omega = 4.16A$$

$$I_3 (50\Omega) = 125V \div 50\Omega = 2.5A$$

iv) Calculate voltage drop for each resistor

Circuit a

$$V = IR$$

$$V_1 = 1.25A \times 20\Omega = 25V$$

$$V_2 = 1.25A \times 30\Omega = 37.5V$$

$$V_3 = 1.25A \times 50\Omega = 62.5V$$

Circuit b

Voltage same across resistor

$$V_1 = V_2 = V_3 = 25V$$

v) Calculate power dissipated through each resistor

$$P_1 = 25V \times 1.25A = 31.25W$$

$$P_2 = 37.5V \times 1.25A = 46.875W$$

$$P_3 = 62.5V \times 1.25A = 78.125W$$

Circuit b.

$$P_1 = 125V \times 6.25A = 781.25W$$

$$P_2 = 125V \times 4.16A = 520W$$

$$P_3 = 125V \times 2.5A = 312.5W$$